# REPORT OF INTERNSHIP TRAINING

# AT

# SOFTRONIICS

**From 16/04/2026 to 16/05/2026**

Submitted for partial fulfillment of the requirements of awarding Degree in

**B. Voc Data Science and Analytics**

**University of Calicut**


Submitted By

**SREELAKSHMI.V**

**POAYBOE026**

2024-2027

Under the guidance of

**Mrs. Shirin Shahnas. B. V**

Assistant Professor

Dept of Data Science





**Department of Data Science and Analytics**

**MES PONNANI COLLEGE, PONNANI**

# MES PONNANI COLLEGE

NAAC Re-accredited (III Cycle) with A+ Grade (3.46 CGPA) | Recognized under UGC 2(f) & 12(B)
Govt. Aided Institution | Affiliated to the University of Calicut | ISO 9001:2015 Certified | Estd.: 1968

## CERTIFICATE

### Department of Data Science and Analytics

This is to certify that **SREELAKSHMI. V**, Registration Number **POAYBOE026**, student of Bachelor of Vocational Education in Data Science and Analytics 2024-2027 Batch, has successfully completed the internship program at **SOFTRONIICS** from 16/04/2026 to 16/05/2026 as part of the curriculum requirements.

The department has reviewed the student's internship report and the evaluation from the industry supervisor, and confirms that the internship experience meets the academic standards and learning objectives of the B Voc Data Science program.

**Principal** **Head of the Department** **Faculty Supervisor**

**Valued on:** ……

**Examiners:**

**Register No:** …….....

1. …….

**Name**: ……..

2. …….

# ORGANIZATION CERTIFICATE

Softroniics®

# CERTIFICATE OF INTERNSHIP

THIS IS TO CERTIFY THAT

SREELAKSHMI V

REG No. POAYBOE026

Student of *MES PONNANI COLLEGE* pursuing *2nd YEAR, B.VOC Data Science & Analytics* has successfully completed 60 hours of internship from 16/04/2026 to 16/05/2026 in *AI/ML*. We wish good luck for future endeavors.

Softroniics Pvt Ltd Maniyattukudi Building Calicut

CERTIFIED ISO 9001:2015 CERTIFIED

MANIKANDAN.P
CEO SOFTRONIICS

# DECLARATION

I **SREELAKSHMI. V, POAYBOE026** hereby declare that this internship report, Submitted to Dept of Data Science Mes Ponnani College, in partial fulfillment of the requirements for the degree of B. Voc in Data Science and Analytics, University of Calicut is an original work done by me on the period of date with the support of **Softroniics**. All sources of information and assistance received during the internship period have been duly acknowledged.

Name : **SREELAKSHMI. V**

Signature:

Place :

Date :

# ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported me throughout my internship experience at **SOFTRONIICS**. This opportunity has been invaluable for my professional and personal development in the field of data science, Artificial Intelligence and Machine Learning.

First and foremost, I am deeply grateful to my industry supervisor, NAILA U.U, for their expert guidance, continuous support, and willingness to share their wealth of knowledge and experience. Their mentorship has significantly enhanced my understanding of real-world data science applications and professional practices.

I extend my heartfelt thanks to my faculty supervisor, SHIRIN SHAHNAS B.V, HOD ASSISTANT PROFESSOR Department of Data Science for her academic guidance, constructive feedback, and encouragement throughout this journey. Their insights helped me bridge theoretical knowledge with practical implementation.

I wish to express my sincere thanks to the Department of Data Science at MES PONNANI COLLEGE for providing this invaluable opportunity as part of the curriculum. The knowledge and skills acquired during my coursework served as a strong foundation for this practical experience.

**This internship experience has been transformative, providing me with practical insights and reinforcing my passion for data science. The skills and knowledge gained will undoubtedly help shape my future career path.**

**SREELAKSHMI. V**

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# EXECUTIVE SUMMARY

This report documents my internship experience at Softroniics, Calicut, from April 16, 2026 to May 16, 2026, as part of the B. Voc Data Science and Analytics program at MES Ponnani College. The internship was focused on *implementing data analytics and preprocessing solutions for health risk prediction within the healthcare domain.*

During the internship, I was integrated into the company's data science team and worked on *developing clinical insights and preprocessing data pipelines that analyse patient health characteristics to improve stroke risk prediction effectiveness. The primary project involved building a robust exploratory data analysis framework, handling missing physiological metrics, and developing a structured feature engineering pipeline to convert clinical variables into machine-learning-ready formats. Additionally, I contributed to optimizing the data pipeline by addressing categorical data structures and implementing custom visualizations for critical binary medical indicators.*

The technical scope of the Internship encompassed working with *NumPy, Pandas, Matplotlib, Seaborn, Git, GitHub, and Streamlit*. I applied various data science methodologies including *exploratory data analysis, grouping, merging, and joining data structures, missing data imputation, outlier analysis, and feature scaling*. The business context required understanding *healthcare metrics, clinical risk factors, classification machine learning algorithms, and model optimization techniques.*

Key achievements during the internship include:

- *Development of an end-to-end data preprocessing pipeline using Pandas and NumPy that handles train-test splitting, data cleaning, text encoding, and feature scaling while mitigating outlier distortion.*
- *Implementation and evaluation of multiple classification models—including Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Machines (SVM), Decision Trees, and Ensemble methods—while addressing overfitting, underfitting, and regularization.*
- *Deployment of the finalized, hyperparameter-tuned model into a functional web application interface using Streamlit, using version control through Git and GitHub to manage code deployments.*

The internship provided valuable industry exposure and helped bridge the gap between academic knowledge and real-world applications of data science. It enhanced my technical skills in machine learning, data engineering, and business analytics while also developing professional competencies in project management, communication, and teamwork.

This report details the projects undertaken, methodologies applied, challenges faced, solutions implemented, and key learnings gained throughout the internship period. It concludes with reflections on how this experience has shaped my understanding of the data science field and influenced my career aspirations.

# 1. INTRODUCTION

## 1.1 About the Company

Softroniics is a software development and professional training organization with more than 15 years of experience in delivering innovative technology solutions and industry-focused skill development programs. Over the years, we have successfully worked with clients across various sectors while helping students and professionals build successful careers through practical training and industry exposure. We are committed to bridging the gap between academic education and industry requirements by offering a combination of software development services, industry-oriented training, internships, live projects, and placement support

**Our Expertise**:

● 15+ Years of Experience in software development and technology training.

●Technology Consulting and Digital Solutions tailored to business requirements.

● Industry-Oriented Training: Specialized programs designed to equip students with current and in-demand technical skills.

● Software Development Services: End-to-end software solutions and client projects that provide practical exposure to real-world business requirements.

● Industry-Ready Programs: Structured learning pathways that develop technical, communication, aptitude, and professional workplace skills.

● Internships and Live Projects: Opportunities for students to work on practical assignments and gain valuable industry experience.

● Placement Assistance: Career guidance and placement support through collaborations with organizations across multiple industries.

**Our Vision**

To become a leading technology and training organization that empowers students, professionals, and businesses through innovation, practical learning, and industry-driven solutions.

**Our Mission**

To develop skilled and employable talent by providing quality training, real-world project experience, and career opportunities while delivering reliable software solutions to clients worldwide.

**Our Digital Platforms**

● www.softroniics.in – Dedicated to software development services, client projects, and business technology solutions.

● www.softroniics.com – Dedicated to training programs, internships, industry-ready initiatives, placements, and career development services.

With a strong foundation built over 15+ years, Softroniics continues to support educational institutions, businesses, and aspiring professionals by delivering excellence in technology, training, and career advancement.

We would be delighted to collaborate with esteemed institutions in conducting training programs, internships, workshops, placement initiatives, and industry interaction activities that contribute to the professional growth of students.

# ORGANIZATIONAL STRUCTURE




## 1.2 Departments and their nature of work

During my internship, I was assigned to the Data Science and Analysis team within the Operations department at Softroniics. The department acts as the primary technical hub of the organization, managing live project engineering alongside advanced training workflows. The complete hierarchy and operational breakdown of the company units include:

- **CEO** — sits at the top of the hierarchy and holds overall strategic authority. The Sales Head and Marketing Head both Report directly to the CEO, keeping revenue-generating functions close to leadership.
- **COO** — reports to the CEO and oversees day-to-day operations. The Operations/Branch Head reports directly to the COO.

- **Sales** — headed by the Sales Head who reports directly to the CEO. The team includes Sales Executives who execute client acquisition, lead follow-ups, field work, and core revenue targets.
- **Marketing** — headed by the Marketing Head who reports directly to the CEO. The department has two wings: Digital Marketing (online campaigns, SEO, social media) and Direct Marketing (offline outreach, events, direct communications).
- **Operations** — led by the Operations/Branch Head who reports directly to the COO. This department is the backbone of delivery — ensuring quality project delivery to clients, managing student training programs, and supporting trained students through placement. The department is divided into four specialized functional arms:
- **Research & Development** — reports to the Operations Head. The R&D team focuses on new technology research, product development, and innovation/process improvement, keeping the organization ahead of the curve in its service and solution offerings.
- **HR** — led by the HR Head, reporting to the Operations Head. Responsibilities cover talent acquisition (hiring, onboarding, recruitment), people management, employee engagement/development, and HR policies/compliance.
- **Finance** — led by the Finance Head, reporting directly to the Operations Head. The team handles financial planning and budgeting, core accounting/bookkeeping, compliance/reporting, and cost control analysis to keep fiscal management integrated with operational needs.

## 1.3 Internship Objectives

### 1.3.1 Academic Objectives:

- Apply Theoretical Knowledge: Apply core data science, machine learning, and statistical analysis concepts learned throughout the B. Voc Data Science and Analytics program to high-impact, real-world healthcare predictive modelling.

- Master the End-to-End Workflow: Gain intensive, practical experience across the complete data lifecycle—ranging from raw exploratory data analysis (EDA) and data cleansing to feature engineering, model optimization, and production-ready deployment.

- Tool and Technology Proficiency: Develop advanced proficiency in industry-standard technologies and libraries utilized in commercial data engineering, specifically Python, Scikit-Learn, Pandas, NumPy, and Streamlit.

- Academic Integration: Produce robustly documented code, pipeline binaries, and analytical evidence of applied learning to successfully fulfill the institutional curriculum and internship requirements of the degree program.

### 1.3.2 Company-Assigned Objectives:

- End-to-End Diagnostic Pipeline Development: Participate in data preparation, handling missing values, and executing feature engineering pipelines on complex multi-type healthcare datasets.
- Multi-Model Evaluation & Optimization: Train, cross-validate, and benchmarking multiple machine learning frameworks (including Random Forest, Decision Trees, KNN, AdaBoost, and Logistic Regression) to isolate the most reliable classification engine.
- Performance Optimization for Clinical Risk: Evaluate models using confusion matrices to optimize classification thresholds, focusing strictly on minimizing False Negatives to ensure patient diagnostic safety.
- Interactive Dashboard Deployment: Design, code, and deploy an interactive, user-facing web application using Streamlit to allow healthcare providers to run real-time, automated client risk assessments.
- Technical Documentation & Reporting: Support the engineering team by generating structured performance reports, system architecture mapping, and visualization summaries for stakeholders.

### 1.3.3 Personal Learning Objectives:

- Advanced Machine Learning Practice: Develop deeper technical expertise in ensemble learning methods (like Random Forest and AdaBoost) used for complex medical and behavioural risk prediction.
- Pipeline Serialization & Architecture: Master best practices in model serialization and deployment workflows by effectively caching preprocessing transformers (LabelEncoder, StandardScaler) and predictive models via pickle.
- Enhance Engineering Soft Skills: Improve project management, problem-solving, and clean-coding habits while translating abstract academic concepts into fully functional software products.

These objectives were designed to be measurable and achievable within the internship period while ensuring alignment with both academic requirements and Softroniics' business needs. Regular check-ins with both my faculty supervisor and industry mentor helped track progress against these objectives and make necessary adjustments throughout the internship

# 2. TECHNICAL BACKGROUND

## 2.1 Relevant Technologies and Tools

During my internship at Softroniics, Calicut, I gained practical exposure to various technologies and tools used in the fields of python programming, data science, artificial intelligence, and machine learning. Under the guidance of my mentor, I learned and implemented these technologies while developing an end-to-end machine learning stroke prediction pipeline. The key technologies and tools used during the internship are summarized below:

- **Programming languages**:
  python.
- **Data processing and analysis**:
  numpy,
  pandas,
  data cleaning techniques,
  and data manipulation using group,
  merge,
  and join operations.
- **Data preprocessing techniques**:
  train-test split,
  feature encoding,
  feature scaling,
  outlier analysis,
  boxplot analysis for outlier detection,
  and imbalanced data handling via smote (synthetic minority oversampling technique).
- **Data visualization**:
  matplotlib and seaborn.
- **Machine learning and artificial intelligence**:
  scikit-learn,
  linear regression,
  logistic regression,
  k-nearest neighbors (knn),

decision tree,

random forest,

support vector machine (svm),

and ensemble learning methods.

- **Model evaluation and optimization**:

  cross-validation,

  hyperparameter tuning,

  overfitting and underfitting analysis,

  and regularization techniques.

- **Development and deployment tools**:

  visual studio code (vs code) and streamlit.

- **Version control and collaboration**:

  git and github.

The combination of these technologies and tools provided a comprehensive learning environment for understanding the end-to-end machine learning development lifecycle. The internship enabled me to gain practical experience in data preprocessing, exploratory data analysis, machine learning model development, model evaluation, and deployment through a streamlit-based application. This hands-on exposure significantly enhanced my technical knowledge and practical skills in artificial intelligence and machine learning.

# 3. INTERNSHIP WORK AND RESPONSIBILITIES

## 3.1 Overview of Assigned Tasks

During my internship at Softroniics from 16 April 2026 to 16 May 2026, I participated in a structured Artificial Intelligence and Machine Learning training program. The internship focused on developing practical skills in data science, machine learning, data visualization, model development, and deployment techniques through hands-on exercises, assignments, and guided learning sessions.

The internship activities were organized in a progressive manner, allowing me to build a strong foundation in data analysis and machine learning concepts while directly executing an end-to-end medical risk analytics project: **The Stroke Prediction System**.

**Week 1: Environment Setup & Data Handling**

- Attended project orientation and gained an understanding of project objectives, workflow, and expected outcomes.
- Set up and configured the development environment using Visual Studio Code (VS Code), Python, and required libraries.
- Learned the fundamentals of Python programming for data analysis and machine learning applications.
- Explored NumPy arrays and performed efficient vectorized mathematical operations on large datasets.
- Worked with multidimensional arrays, indexing, slicing, reshaping, and aggregation functions.
- Gained hands-on experience with Pandas DataFrames and Series for structured data management.
- Imported healthcare datasets from CSV files and performed initial data inspection.
- Conducted basic data cleaning tasks, including handling missing values and removing duplicate records.
- Understood different data types and data structures used in data analysis.

**Week 2: Data Manipulation & Visualization**

- Performed advanced data manipulation using Pandas, including filtering, sorting, grouping, and aggregation operations.
- Merged and joined multiple data features to create comprehensive data tables for analysis.
- Conducted exploratory data analysis (EDA) to identify trends, relationships, and medical risk factor inconsistencies in data.
- Generated descriptive statistical summaries to understand clinical dataset characteristics.
- Created line charts, bar charts, histograms, and scatter plots using Matplotlib.

- Customized visualizations with labels, legends, colours, and grid formatting for better presentation.
- Used Seaborn to develop advanced statistical visualizations such as boxplots, heatmaps, and pair plots.
- Identified patient parameter outliers and analysed data distributions using visualization techniques.
- Learned best practices for presenting analytical findings through visual storytelling.

**Week 3: Machine Learning Fundamentals & Preprocessing**

- Studied core machine learning concepts, including supervised and unsupervised learning approaches.
- Learned the workflow of a machine learning project from data collection to model evaluation.
- Performed data preprocessing to improve data quality and model readiness.
- Detected and handled clinical data outliers using statistical methods and visualization techniques.
- Managed missing values through appropriate imputation methods to handle features like body mass index.
- Applied train-test split techniques to evaluate model generalization performance.
- Implemented categorical feature encoding using Label Encoding and One-Hot Encoding methods.
- Performed feature scaling using Standardization and Normalization techniques.
- Understood feature selection concepts and their impact on model performance.
- Explored evaluation metrics for classification and regression tasks.

**Week 4: Model Building & Evaluation**

- Built baseline predictive models using Linear Regression for continuous target prediction benchmarking.
- Implemented Logistic Regression models for stroke classification problems.
- Studied the concepts of overfitting and underfitting and their impact on model performance.
- Developed K-Nearest Neighbors (KNN) classification models and analysed parameter sensitivity.
- Built Support Vector Machine (SVM) models and explored different kernel functions.
- Implemented Decision Tree models and interpreted decision paths.
- Compared the performance of multiple machine learning algorithms using evaluation metrics.
- Generated confusion matrices and classification reports to assess classification accuracy.

**Week 5: Advanced Models, Deployment & Version Control**

- Implemented ensemble learning techniques using Random Forest classifiers.
- Addressed class imbalance problems within the clinical target features using SMOTE (Synthetic Minority Over-sampling Technique).
- Performed cross-validation to obtain reliable model performance estimates.
- Conducted hyperparameter tuning using Grid Search and Random Search techniques.

- Optimized model accuracy, precision, recall, and F1-score through parameter adjustments.
- Developed an interactive web application using Streamlit for model deployment.
- Integrated trained machine learning models into the deployment application.
- Tested and validated the application under different user input scenarios.
- Managed source code and project versions using Git and GitHub.
- Learned collaborative development practices, including branching, commits, and repository management.
- Documented project workflows and deployment procedures.
- Successfully demonstrated and presented the final machine learning stroke prediction solution to the industry mentor and project team.

## 3.2 Project Contribution

During the internship at Softroniics, significant contributions were made to the development and deployment of the automated stroke prediction machine learning system:

- End-to-end pipeline engineering: built a comprehensive data pipeline from scratch using python, transitioning seamlessly from data ingestion to a functional local web interface.
- Data integrity & preprocessing: executed core data cleaning protocols including data imputation strategies for missing metrics, categorical encoding, and feature scaling to ensure model accuracy.
- Exploratory data analysis (EDA): engineered clear, structured statistical charts and heatmaps using matplotlib and seaborn to visually isolate core physiological stroke risks.
- Predictive model development: implemented, optimized, and compared multiple machine learning algorithms—ranging from baseline regressions to complex ensemble architectures.
- Model optimization & validation: applied robust cross-validation methods and detailed hyperparameter tuning to mitigate model overfitting and improve predictive safety benchmarks.
- Interactive system deployment: designed and built a lightweight web dashboard using the streamlit framework, turning complex predictive math into an accessible portal for mock clinical risk evaluation.

## 3.3 Challenges and Solutions

Throughout my internship at Softroniics, I encountered various challenges that required creative problem-solving and technical expertise. These challenges provided valuable learning opportunities and helped me develop both technical and professional skills. Below are the key challenges I faced and the solutions I implemented:

**Challenge 1:**

**Handling Mixed Data Types and Maintaining Data Pipeline Integrity**

- **Problem:**

  The raw healthcare dataset (healthcare-dataset-stroke-data.csv) contained a mix of continuous numbers (e.g., age, glucose levels, BMI) and text-based categorical fields (e.g., gender, smoking status, work type). Early iterations of the model pipeline crashed during training because the preprocessing steps threw errors when handling strings, and missing values in the bmi column skewed the arrays. Furthermore, failing to preserve the exact categorical transformations between the training notebook and the deployment script caused the system to misinterpret user inputs.

- **Solution:**

  I built a systematic data engineering pipeline in my cleaning.ipynb notebook using Scikit-Learn. I isolated every categorical column and fit five separate LabelEncoder objects (le, le1, le2, le3, le4) to map text labels to uniform integers. I also applied a StandardScaler (q) to center and scale the numeric features. Crucially, I used Python's pickle library to freeze and save all seven individual transformation components as.save binary files (encoded_gender.save, stroke_scalar.save, etc.) so they could be identically reloaded in production.

- **Outcome:**

  This completely eliminated datatype conversion runtime crashes. It ensured that the data distribution seen by the model during the training phase perfectly mirrored the live inputs submitted through the web interface, preserving mathematical integrity across the entire project.

**Challenge 2:**

**Optimizing Model Selection for High-Risk Clinical Safety**

- **Problem:**

  When benchmarking various machine learning classifiers, standard accuracy scores did not reflect the actual safety needs of a clinical environment. For example, some models achieved a decent overall accuracy but generated an uncomfortably high rate of False Negatives

  (missing up to 296 true stroke cases in weaker test runs). In medical diagnostics, a False Negative—telling a high-risk patient they are perfectly healthy—is a catastrophic failure that can delay life-saving triage.

- **Solution:**

  I implemented a Stratified 3-Fold Cross-Validation pipeline to evaluate five different model frameworks: Random Forest, Decision Trees, KNN, AdaBoost, and Logistic Regression. Instead of looking only at the overall mean accuracy, I generated and deeply analysed a structural confusion matrix for each algorithm to track their exact classification behaviours on a consistent test pool of 1,945 samples.

- **Outcome:**

  This structural analysis revealed that the RandomForestClassifier was the superior choice. It achieved the highest mean validation accuracy (**93.73%**) while drastically minimizing False Negatives down to just 90 missed cases compared to its peers. This ensured the final deployed engine was optimized for true clinical utility and diagnostic safety.

**Challenge 3:**

**Transitioning Core Code Blocks to a Stateful Front-End UI**

- **Problem:**

  During the initial build of the interactive Streamlit deployment application (stroke.py), the program structure was broken by several syntax errors and data type mismatches. Streamlit checkboxes (used for hypertension and heart disease history) naturally returned Python boolean values (True/False), whereas the model scaler expected binary integers (1/0). Additionally, using standard st.text_input fields returned raw text strings for age, glucose, and BMI, which broke the backend mathematical array operations. Finally, an indentation error inside the script's main loop prevented the framework from initializing altogether.

- **Solution:**

  I completely overhauled the structure of stroke.py. I replaced the textual entry fields with explicit st.number_input widgets to enforce numeric data entry boundaries from the interface level. I implemented inline conditional casting (1 if ... else 0) on top of the checkboxes to instantly map user clicks to valid binary machine learning features. Lastly, I refactored the script execution layout by un-indenting the core call and nesting it cleanly under a standard Python entry baseline block: if __name__ == '__main__': main().

- **Outcome:**

The user interface transitioned into a highly stable, production-ready graphical application. The interactive dashboard successfully processes real-time clinical metrics and generates instant, bug-free, color-coded visual alert banners (st.success and st.error) depending on the patient's predicted risk profile.

# WORK DIARY

**INTERNSHIP WEEKLY REPORT – 1st week**

Student Name : _____Sreelakshmi V _____

Week Beginning : ___16/04/2026__To___18/04/2026_____

Company Name : _____Softroniics _____

Position : ______Intern_____

Internship Supervisor's Name: _____Naila U U _____Phone: ___8921479099 ____

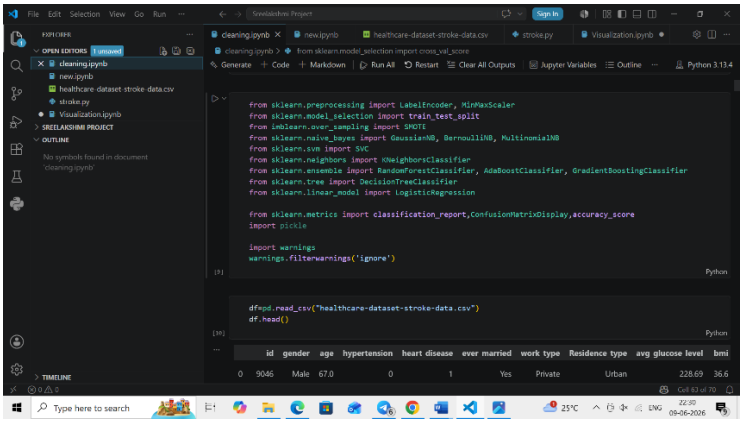
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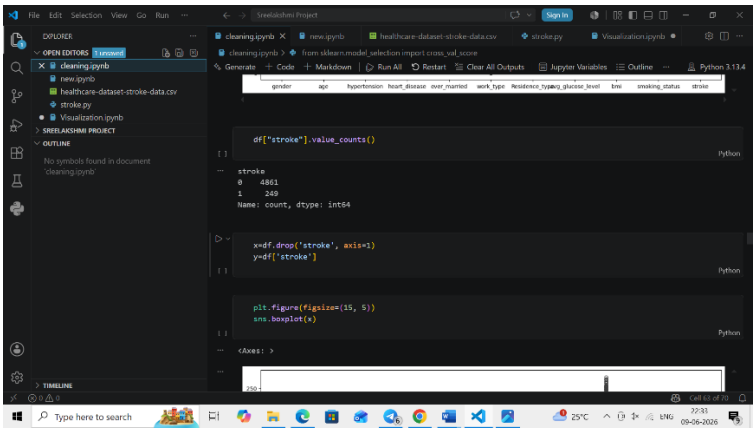
| DAY | TIME IN | TIME OUT | TOTAL HOURS | MAJOR DUTIES PERFORMED |
|---|---|---|---|---|
| MONDAY | — | — | — | — |
| TUSEDAY | — | — | — | — |
| WEDNESDAY | — | — | — | — |
| THURSDAY | 11.30 AM | 2.30 PM | 3 | Studied the fundamentals of NumPy, focusing on array creations, slicing, and vectorization operations. |
| FRIDAY | — | — | — | — |
| SATURDAY | 11.30 AM | 2.30 PM | 3 | Performed data cleaning, handling missing values, and preprocessing using the Pandas library. |
| SUNDAY | — | — | — | — |
| **TOTAL HOURS** | | | 6 | |

Signature: …… ……

Student Intern Internship Supervisor

Screenshots: -



**INTERNSHIP WEEKLY REPORT – 2nd week**

Student Name : _____Sreelakshmi V _____

Week Beginning : ___21/04/2026____ To ___25/04/2026___

Company Name : _____Softroniics _____

Position : _____Intern_____

Internship Supervisor's Name: _____Naila U U _____Phone: ___ 8921479099 ____

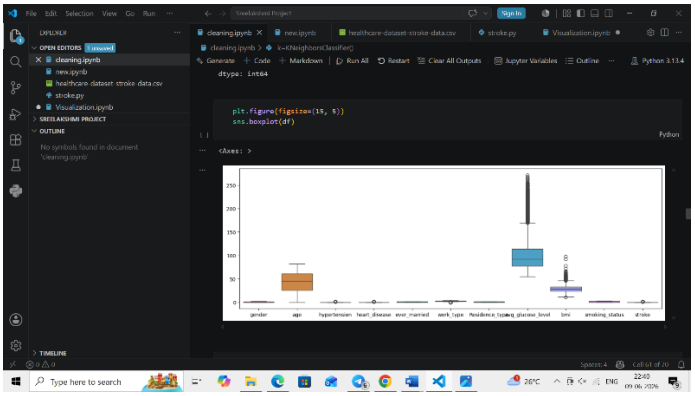
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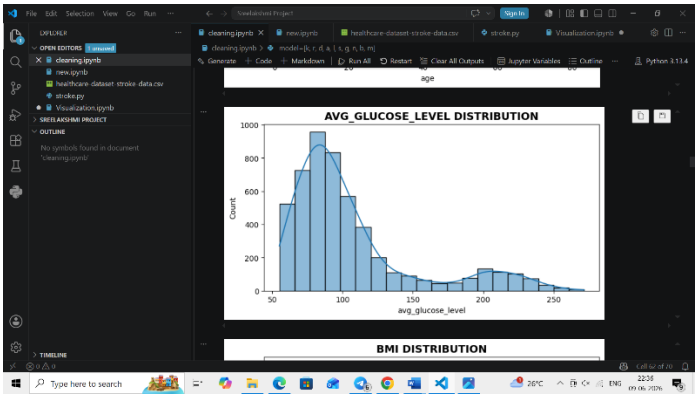
| DAY | TIME IN | TIME OUT | TOTAL HOURS | MAJOR DUTIES PERFORMED |
|---|---|---|---|---|
| MONDAY | ___ | ___ | ___ | ____ |
| TUSEDAY | 9.30 AM | 2.30 PM | 5 | Applied group, merge, and join operations in Pandas to organize and restructure the dataset features. |
| WEDNESDAY | | | | |
| THURSDAY | 10.30 AM | 2.30 PM | 4 | Created exploratory data visualizations and basic feature distribution charts using Matplotlib. |
| FRIDAY | | | | |
| SATURDAY | 9.30 AM | 2.30 PM | 5 | Developed correlation heatmaps and patient risk factor statistical plots using Seaborn. |
| SUNDAY | --- | --- | --- | --- |
| **TOTAL HOURS** | | | 14 | |

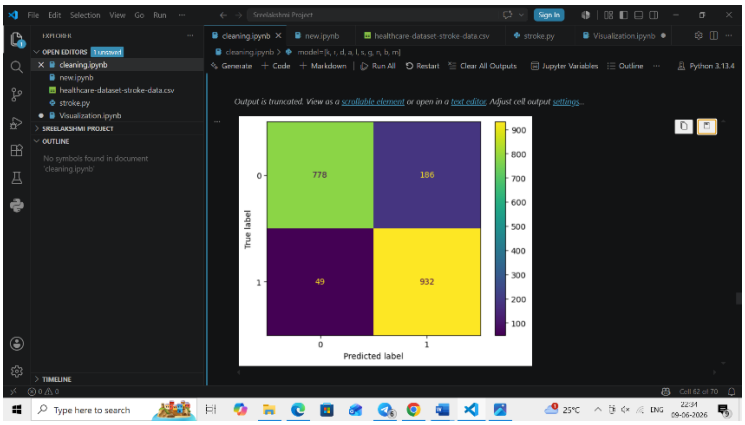
Signature: …… ……..

Student Intern Internship Supervisor

Screen shots: -





**INTERNSHIP WEEKLY REPORT – 3 rd week**

Student Name : _____Sreelakshmi V _____

Week Beginning : ___29/04/2026____ To ___01/05/2026___

Company Name : _____Softroniics _____

Position : _____Intern_____

Internship Supervisor's Name : _____Naila U U _____ Phone: ___ 8921479099____

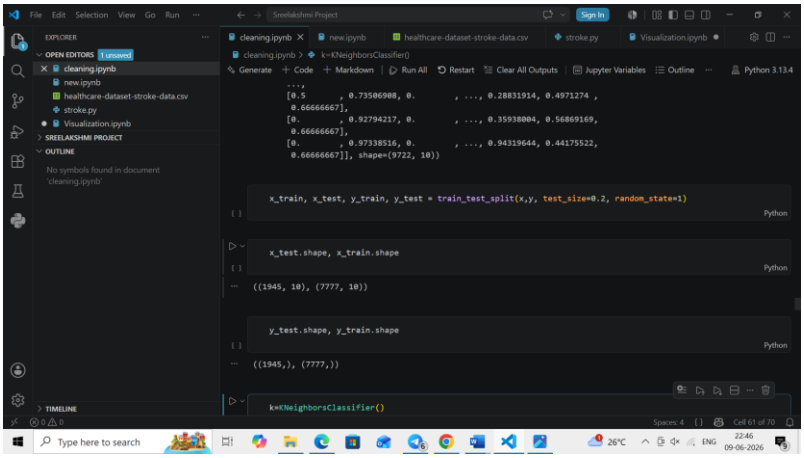
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| DAY | TIME IN | TIME OUT | TOTAL HOURS | MAJOR DUTIES PERFORMED |
|---|---|---|---|---|
| MONDAY | ___ | ___ | ___ | ___ |
| TUSEDAY | | | | |
| WEDNESDAY | 9.30 AM | 2.30 PM | 5 | Learned machine learning fundamentals, classification vs. regression concepts, and explored Linear and Logistic Regression baselines. |
| THURSDAY | ___ | ___ | ___ | ___ |
| FRIDAY | 9.30 AM | 2.30 PM | 5 | Applied train-test split, label encoding, standard feature scaling, outlier analysis, and studied overfitting, underfitting, and regularization. |
| SATURDAY | ___ | ___ | ___ | ___ |
| SUNDAY | ___ | ___ | ___ | ___ |
| **TOTAL HOURS** | | | 10 | |

Signature: …… ……

Student Intern Internship Supervisor

Screen shots: -



**INTERNSHIP WEEKLY REPORT- 4 th week**

Student Name : _____Sreelakshmi V _____

Week Beginning : ___04/05/2026____ To ___08/05/2026___

Company Name : _____Softroniics _____

Position : _____Intern_____

Internship Supervisor's Name : _____Naila U U _____ Phone: ___ 8921479099 ____

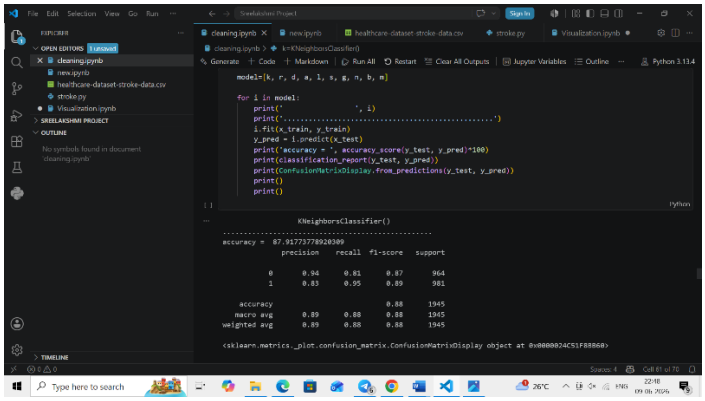
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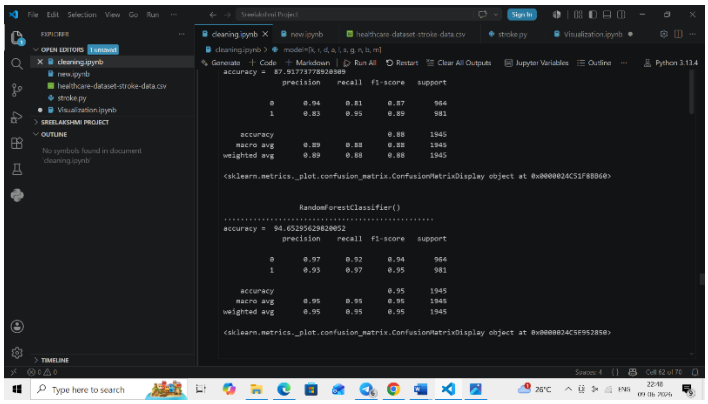
| DAY | TIME IN | TIME OUT | TOTAL HOURS | MAJOR DUTIES PERFORMED |
|---|---|---|---|---|
| MONDAY | 9.30 AM | 2.30 PM | 5 | Studied the core concepts of model overfitting, underfitting, and variance regularization techniques. |
| TUSEDAY | ----- | ----- | ------ | ----- |
| WEDNESDAY | 9.30 AM | 2.30 PM | 5 | Implemented K-Nearest Neighbors (KNN), Support Vector Machine (SVM) algorithms, and built Decision Tree models. |
| THURSDAY | ---- | ---- | ---- | ---- |
| FRIDAY | 9.30 AM | 2.30 PM | 5 | Analyzed and understood Ensemble Learning frameworks to enhance final project predictive performance. |
| SATURDAY | ---- | ---- | ---- | ----- |
| SUNDAY | ---- | ---- | ---- | ---- |
| **TOTAL HOURS** | | | 15 | |

Signature: …… ……

Student Intern Internship Supervisor

Screenshots: -



**INTERNSHIP WEEKLY REPORT – 5 th week**

Student Name : _____Sreelakshmi V _____

Week Beginning : ___11/05/2026____ To ___15/05/2026___

Company Name : _____Softroniics _____

Position : _____Intern_____

Internship Supervisor's Name : _____Naila U U _____ Phone: ___ 8921479099 ____

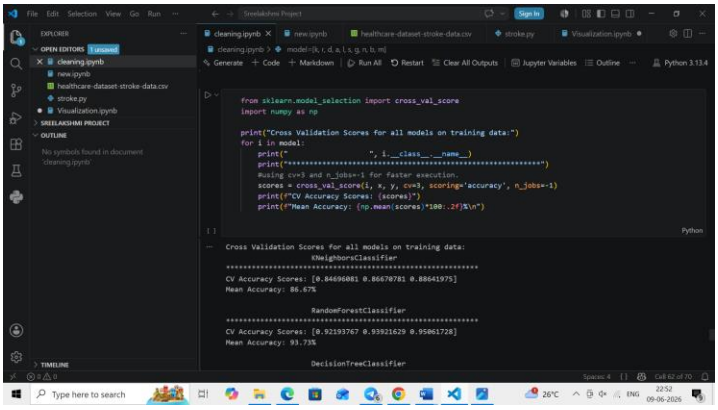
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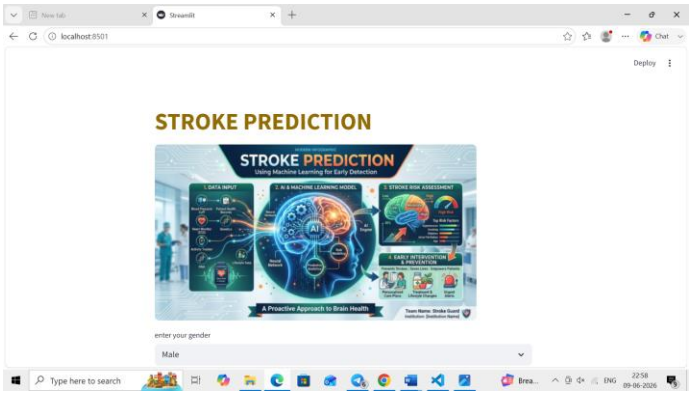
| DAY | TIME IN | TIME OUT | TOTAL HOURS | MAJOR DUTIES PERFORMED |
|---|---|---|---|---|
| MONDAY | 9.30 AM | 2.30 PM | 5 | Applied Cross-Validation methods and performed Hyperparameter Tuning to improve stroke prediction model performance. |
| TUSEDAY | ----- | ---- | --- | ---- |
| WEDNESDAY | 9.30 AM | 2.30 PM | 5 | Developed the web interface and implemented live model deployment using the Streamlit framework. |
| THURSDAY | --- | --- | --- | --- |
| FRIDAY | 9.30 AM | 2.30 PM | 5 | Managed version control using Git and GitHub, finalized documentation, and attended soft skill development sessions. |
| SATURDAY | --- | --- | --- | --- |
| SUNDAY | --- | --- | --- | --- |
| **TOTAL HOURS** | | | 15 | |

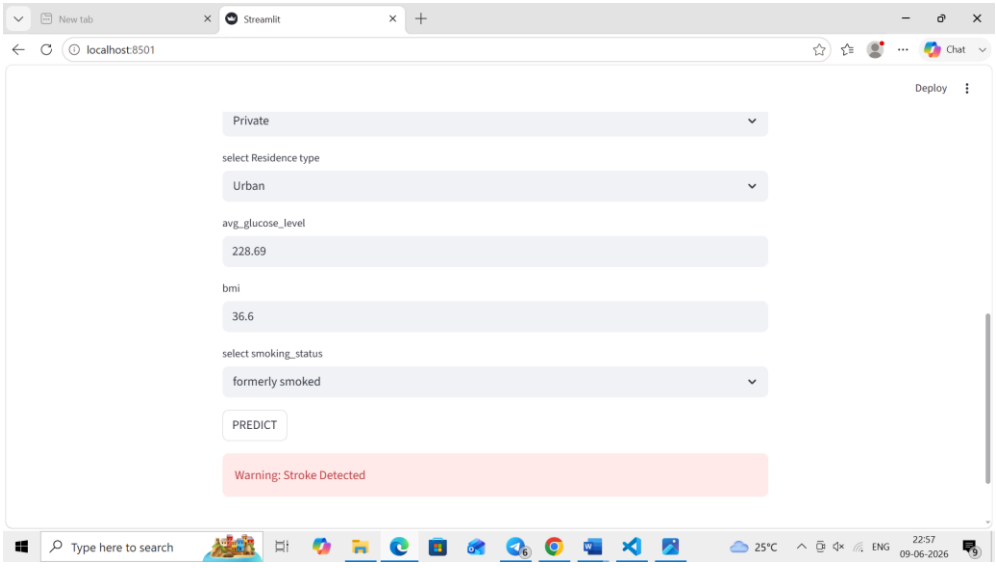
Signature: …… ……

Student Intern Internship Supervisor

Screenshots: -



# 4. SWOT ANALYSIS

The overall evaluation of a company's strengths, weaknesses, opportunities, and threats is called SWOT analysis. In a SWOT analysis, the best strategies accomplish an organization's mission by exploiting an organization's opportunities and strengths, neutralizing its threats, and avoiding or correcting its weaknesses.

**STRENGTHS**

- **Structured Technical Mentorship:** Softronics provides dedicated, direct expert mentorship that bridges the gap between academic theory and practical code implementation.
- **Core Technological Focus:** The training curriculum specializes heavily in highly sought-after industry tools like Python, VS Code, and advanced AI/ML libraries.
- **Production-Oriented Capstone Training:** The program emphasizes practical development, pushing student concepts from raw datasets directly into live Streamlit web deployment.
- **Optimized Learning Environment:** Softronics delivers a streamlined, distraction-free environment tailored for rapid technical skill development.

**WEAKNESSES**

- **Limited Interaction with Corporate Peer Teams:** The program focuses heavily on independent project execution under a single mentor, limiting exposure to multi-department corporate structures.
- **Absence of Live Production Data Workflows:** Training is conducted on curated reference datasets rather than unpredictable, high-velocity commercial client streams.

**OPPORTUNITIES**

- **Industry Networking Links:** Leveraging the company's regional position in Calicut to connect with technical communities and explore professional data roles.
- **Hands-on Portfolio Expansion:** Utilizing the deployed Streamlit predictive app as a foundational proof-of-work project for prospective employment opportunities.

**THREATS**

- **Rapidly Evolving AI Framework Ecosystems:** The fast pace of changes in machine learning libraries requires continuous post-internship upskilling to maintain technological relevance.
- **Highly Competitive Junior Data Market:** Navigating an entry-level professional field that increasingly demands extensive multi-tool fluencies alongside core programming skills.

# 5. FUTURE RECOMMENDATIONS

- **Extend Program Duration:** Extend the duration of the internship program to provide deeper practical exposure to advanced Artificial Intelligence, Machine Learning, and Data Science applications.
- **Integrate Production-Scale Clinical Data:** Provide opportunities to work with larger, more diverse, and real-world production datasets to enhance analytical and clinical problem-solving skills beyond static reference files.
- **Expand Deep Learning Horizons:** Include advanced Machine Learning and Deep Learning topics such as Neural Networks, Natural Language Processing (NLP), Computer Vision, and advanced Gradient Boosting architectures (such as XGBoost and LightGBM).
- **Emphasize Complete Project Lifecycles:** Encourage the development of end-to-end projects covering automated data collection, robust preprocessing pipelines, model development, cloud deployment, and long-term maintenance.
- **Diversify Cloud Deployment Options:** Introduce multi-platform cloud deployment systems such as AWS, Streamlit Community Cloud, or Render to provide diverse experience with real-world application hosting.
- **Incorporate Enterprise Databases:** Provide direct exposure to industry-standard relational database systems such as PostgreSQL and cloud-based data storage infrastructure.
- **Integrate Full-Scale MLOps Practices:** Incorporate specialized MLOps tools and practices, including Git, GitHub, MLflow, and Data Version Control (DVC), to master industry-level model reproducibility and code collaboration.
- **Deepen Optimization Training:** Conduct additional hands-on training dedicated specifically to advanced feature engineering pipelines, complex model optimization, and automated hyperparameter tuning techniques.
- **Foster Collaborative Engineering Teams:** Encourage collaborative development frameworks through team-based peer code reviews and agile project management methodologies.
- **Promote Industry-Readiness Challenges:** Promote participation in hackathons, open-source repositories, and technical workshops on emerging AI technologies to strengthen competitive practical skills.

The internship provided a strong foundation in Python, Data Science, and Machine Learning. Implementing these recommendations can further enhance technical expertise, practical experience, and professional readiness for careers in Artificial Intelligence, Machine Learning, and Data Science.

# 6. CONCLUSION

The internship at Softronics provided an invaluable opportunity to bridge the gap between theoretical data science concepts and practical, real-world software development workflows. Through the design, development, and deployment of an end-to-end machine learning pipeline for stroke prediction, profound technical insights were gained regarding data preprocessing, feature engineering, and model evaluation metrics. Beyond the technical competencies acquired from executing the assigned project tasks, the experience offered significant professional growth through meaningful mentorship and collaborative industry networking. The constructive feedback and industry insights shared by supervisors and peers have provided a solid foundation for entering the competitive data science workforce upon graduation. Ultimately, this engagement successfully cultivated the technical confidence, problem-solving capabilities, and strategic thinking required to excel in the field of Artificial Intelligence and Data Science.

**SREELAKSHMI V**